

# MoCHII User Guide

## Abstract

How to build and run MoCHII (MOnte Carlo for H II regions): requirements, the complete input-parameter reference, output-file formats, worked examples keyed to the regression tests, and the runbook for adding an ion. The physics and validation are documented separately in docs/MoCHII\_physics.pdf; the atomic-data fitting pipeline in docs/MoCHII\_fitting.pdf.

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## 1. Requirements and build

- Intel oneAPI MPI Fortran (mpiifort/mpiifx; GNU mpif90 supported by the Makefile), HDF5 (default prefix /data/opt/hdf5\_intel; override with `make HDF5_PREFIX=...`), CFITSIO.
- The SEDust dust-emission library, self-contained under SEDust/ in this tree. Build it once with `cd SEDust/sed && ./build_lib.sh` (→ SEDust/sed/lib/libsedust.a and its .mod files, which the Makefile links). The SEDust *data* directory is read at run time and resolved automatically relative to the executable (Section 2.10).
- Python 3 with numpy/scipy (h5py, astropy, matplotlib, PyNeb for the test and pipeline scripts).

Build:

```
cd MoCHII
(cd SEDust/sed && ./build_lib.sh)  # on a fresh checkout
make                               # -> MoCHII.x   (always a full clean rebuild)
make DEBUG=1                       # bounds/traceback build
```

The SEDust library only needs the one-time `build_lib.sh` step; afterward `make` links the in-tree SEDust library. The Makefile intentionally performs a full rebuild every time (the MoCafe policy): there is no inter-module dependency tracking, and stale .mod files after editing `define.f90` otherwise cause runtime namelist errors.

Run:

```
mpirun -np 8 ./MoCHII.x input.in
```

MoCHII is MPI-only; the octree and all leaf arrays live in MPI-3 shared memory (one copy on each node).

## 2. Input parameters

All input is one `&parameters` namelist assigning fields of `par`. Parameters not listed keep the defaults shown. The switches compose: `gas_niter = 0` gives a single rates-only pass; `gas_niter > 0` iterates the ionization; `solve_te` adds the thermal balance; `use_metals` the trace-metal registry; `diffuse_field` the explicit diffuse field (forces case A); `refine_front` the I-front re-refinement; `ion_add_dust` the dust coupling.

### 2.1 Photons and RNG

| parameter  | default | meaning                          |
|------------|---------|----------------------------------|
| no_photons | 1e5     | source packets in each iteration |
| no_print   | 1e7     | progress-print stride            |
| iseed      | 0       | RNG seed (0 = from clock/pid)    |

| parameter       | default  | meaning                                                                                                                                                                                                                                                                                                    |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| launch_sequence | 'random' | quasi-random launch: 'sobol' = Owen-scrambled Sobol coordinates for the stellar (and diffuse) packet launch, indexed by the global photon number (MPI-count-independent launch set); transport/scattering stay pseudo-random; prefer $n_{\text{photons}} = 2^m$ . 'random' is the Mersenne Twister default |
| qmc_seed        | 12345    | scramble key (Launch_sequence='sobol'); fixed across the gas iterations; change it for independent replicates (their scatter is the RQMC error estimate)                                                                                                                                                   |

## 2.2 Grid and dust density

| parameter        | default  | meaning                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| grid_type        | 'car'    | 'amr' (octree) or 'car' (single-level Cartesian, leaf list in raster order; no tree and no geometry arrays in memory — cell centers and half-widths are computed analytically from the raster index; refine_front unavailable). 'uniform' is accepted as an alias for 'car'                                                                                                     |
| car_walk         | 'dda'    | car-grid traversal: 'dda' (default) = dedicated incremental Amanatides–Woo walks ( $t_{\text{max}}/t_{\Delta}$ per axis, comparisons and additions only — the fastest car walk); 'amr' = the octree walks with integer index arithmetic replacing the neighbor table, a verification mode that is bit-identical to the single-level octree (grid_type = 'amr')                  |
| amr_file         | ''       | generic AMR leaf file (FITS/HDF5/text): columns x, y, z, level, nH [, metallicity, xHI, ndust]. If empty, a 'car' grid is built from the namelist instead ('amr' requires the file)                                                                                                                                                                                             |
| nx, ny, nz       | 1, 1, 11 | 'car'-from-namelist cell counts (cubic cells required: $2x_{\text{max}}/n_x = 2y_{\text{max}}/n_y = 2z_{\text{max}}/n_z$ )                                                                                                                                                                                                                                                      |
| xmax, ymax, zmax | 1, 1, 1  | 'car'-from-namelist box half-extents [distance_unit]; box is $[-x_{\text{max}}, x_{\text{max}}]$ etc.                                                                                                                                                                                                                                                                           |
| nH_const         | -1       | if $\geq 0$ , override every leaf's $n_{\text{H}}$ with this uniform value [ $\text{cm}^{-3}$ ] (both 'car' and 'amr'); $< 0$ keeps the file/column density                                                                                                                                                                                                                     |
| density_file     | ''       | read the leaf $n_{\text{H}}$ from a uniform 3D density cube onto a 'car' grid (FITS 3D image, or the MoCHII HDF5 image format; NAXIS1/2/3 = $n_x/n_y/n_z$ , values $n_{\text{H}}$ [ $\text{cm}^{-3}$ ]). $n_x/n_y/n_z$ come from the file, the box is xmax/ymax/zmax; an alternative to amr_file. reduce_factor > 1 down-samples; a rmax/rmin cut can mask the cube to a sphere |

| parameter     | default | meaning                                                                                                                                                                        |
|---------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rmax, rmin    | -999, 0 | spherical density cut on leaf centers (both grids): $n_H = 0$ for $r > r_{\max}$ ( $r_{\max} > 0$ ) and for $r < r_{\min}$ ( $r_{\min} > 0$ ; a shell); radii in distance_unit |
| distance_unit | 'kpc'   | 'kpc'/'pc'/'au'/'" (code units)                                                                                                                                                |

| parameter            | default              | meaning                                                                                                                                                                                                                            |
|----------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dust_model           | 'global_dgr'         | leaf grey opacity: 'none', 'global_dgr' ( $n_H c_{\text{ext}} \cdot \text{DGR}$ ), 'from_file' (ndust column), 'laursen09' (static $x_{\text{HI}}$ column), 'laursen09_live' (computed $x_{\text{HI}}$ , refreshed each iteration) |
| cext_dust            | 1.6059e-21           | reference extinction per H [ $\text{cm}^2$ ] at lambda_ref                                                                                                                                                                         |
| DGR, Z_global, Z_ref | 1e-2, 0.0134, 0.0134 | dust-model scalings                                                                                                                                                                                                                |
| f_ion_dust           | 0.01                 | laursen09 survival of dust in ionized gas                                                                                                                                                                                          |
| f_pah, f_ion_pah     | 0, 0                 | PAH share of the reference extinction and its own ionized-gas survival (laursen09_live)                                                                                                                                            |

## 2.3 Ionizing/FUV band and the source

| parameter          | default      | meaning                                                                                                                                                                                                                                                                                                       |
|--------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| use_ion_band       | F            | must be T (the MoCHII transport)                                                                                                                                                                                                                                                                              |
| ion_energy_mode    | 'continuous' | packet-energy transport mode. The default samples the source energy CDF continuously and evaluates cross sections at the sampled energy. Set 'grouped' explicitly only for compatibility/reproduction calculations that use one representative energy per bin                                                 |
| nnu_ion            | 16           | diagnostic ionizing-energy bins in 'continuous' mode (gates use 32); changing it does not change the physical solution. In 'grouped' mode it is the transport/solver bin count                                                                                                                                |
| eion_min, eion_max | 13.598, 100  | ionizing band edges [eV]                                                                                                                                                                                                                                                                                      |
| ion_align_edges    | T            | align diagnostic bin edges to the ionization thresholds. It does not affect continuous-energy transport. In 'grouped' mode it also prevents a transport bin from straddling a threshold; segments are log-uniform and low-priority metal edges are dropped safely when necessary. Set F for a single log grid |

| parameter                    | default | meaning                                                                                                                                                                                                                                                                                |
|------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| add_fuv                      | F       | FUV extension: extra log bins on [efuv_min, eion_min] below the unchanged ionizing bins; transports FUV grain heating and FUV photoionization of low-threshold metals (Mg I, C I, S I, Fe I); luminosity stays the IONIZING luminosity ( $Q_H$ preserved, the FUV part rides on top)   |
| efuv_min                     | 6.0     | FUV lower edge [eV]                                                                                                                                                                                                                                                                    |
| nnu_fuv                      | 8       | FUV bins (16 recommended with metals)                                                                                                                                                                                                                                                  |
| tstar                        | −999    | Planck source temperature [K]                                                                                                                                                                                                                                                          |
| ion_spectrum                 | ''      | global spectrum file (column 1 plus one value column); its column units are set by spectrum_type (§2.5)                                                                                                                                                                                |
| spectrum_type                | 'shape' | column units of every spectrum-file slot (ion_spectrum, src_spectrum_file, ext_spectrum): 'shape' (arbitrary, normalized to the scale — legacy) or a physical type 'per_ev'/'per_hz'/'per_ang'/'per_um' (absolute); see §2.5                                                           |
| luminosity                   | unset   | ionizing BAND luminosity [erg/s] of the (single) source; for a 'shape'/Planck source unset maps to 1.0 (legacy), for a physical-type file unset means DERIVE it from the file integral; with nsource ≥ 2 it becomes the sum of src_lum (or the equal-split total when those are unset) |
| xs_point, ys_point, zs_point | 0       | single-source position (code units, recentered box; nsource=1)                                                                                                                                                                                                                         |

*Multiple internal point sources* (the scalars above are the nsource=1 case):

| parameter           | default | meaning                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nsource             | 1       | number of internal point sources (0–16); 1 keeps the legacy scalars above, ≥ 2 uses the src_* arrays                                                                                                                                                                                                                                                                               |
| src_x, src_y, src_z | 0       | positions of sources 1..nsource (code units)                                                                                                                                                                                                                                                                                                                                       |
| src_lum             | unset   | ionizing luminosities [erg/s] of the sources; set all or none (none = equal split of luminosity); a partial set aborts                                                                                                                                                                                                                                                             |
| src_tstar           | −999    | Planck temperature [K] for each source (multi-source runs only; a single source uses tstar)                                                                                                                                                                                                                                                                                        |
| src_spectrum_file   | ''      | one multi-column spectrum file for all sources: column 1 then one value column for each source (units set by spectrum_type, §2.5); each source is independently normalized so its ionizing segment carries src_lum( <i>i</i> ) (a physical type with src_lum unset is derived from that column); overrides src_tstar; too few columns abort, extra columns are ignored with a note |

*External isotropic field* (composes with the internal sources; ON when `ext_intensity > 0`, an ISRF preset is named, or a physical-type `ext_spectrum` file is given):

| parameter                    | default | meaning                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ext_intensity</code>   | -1      | band-integrated mean intensity $J$ [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] of the external field; the field is ON when $> 0$ and adds to the internal sources; entering ionizing power $L = \pi J A_{\text{surf}}$ . With a physical-type <code>ext_spectrum</code> file or an ISRF preset it is optional: unset = the absolute $J$ is DERIVED from the file/preset, set = the field is RESCALED to this band $J$ (a note is printed) |
| <code>ext_geometry</code>    | 'rec'   | illuminating surface: 'rec' (box faces, face-area-weighted — correct for a non-cubic box) or 'sph' (bounding sphere of radius <code>rmax</code> about the box center; needs <code>rmax &gt; 0</code> )                                                                                                                                                                                                                                              |
| <code>ext_tstar</code>       | -999    | external-field Planck temperature [K]                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>ext_spectrum</code>    | "       | external-field spectrum: a 2-column file (units set by <code>spectrum_type</code> , §2.5; column 2 is an intensity density $J$ ) OR a reserved ISRF preset name 'draine'/'habing'/'mathis' (FUV-only, absolute, needs <code>add_fuv</code> ); priority <code>ext_spectrum &gt; ext_tstar &gt;</code> the global spectrum                                                                                                                            |
| <code>ext_scale</code>       | 1.0     | dimensionless multiplier on an ISRF preset (ignored when <code>ext_intensity &gt; 0</code> , which rescales the preset to a target band $J$ )                                                                                                                                                                                                                                                                                                       |
| <code>source_geometry</code> | 'point' | legacy external-only shorthand: 'external', 'external_rec', or 'external_sph' forces <code>nsource=0</code> with the external field on (needs <code>ext_intensity &gt; 0</code> ); 'point' keeps the composable form; 'slab' is the plane-parallel slab illumination below                                                                                                                                                                          |

## 2.4 Plane-parallel slab illumination

With `source_geometry = 'slab'` (which requires `xy_periodic`, §2.2) the box is an *xy-periodic slab*: the medium varies in  $z$  (uniform in  $x, y$  for a true 1D column with  $n_x = n_y = 1$ , or an arbitrary 3D tile with  $n_x, n_y > 1$  repeated periodically), and light enters through the top (+ $z$ ) and/or bottom (- $z$ ) face. Each face is either a collimated *beam* at incidence angle  $\theta$  or an *isotropic* (Lambert) field, with its own strength. The single physical normalization is the flux  $F_z$  crossing the (horizontal) illuminated face: for a beam  $F_z = \mu F_n$  ( $\mu = \cos \theta$ ,  $F_n$  the flux normal to the beam), for the isotropic mode  $F_z = \pi I$  ( $I$  the specific intensity).  $F_z$  sets the packet weight through  $L = F_z A_{z\text{face}}$ ; every per-volume result is independent of the tile area. Point-observer peel-off is disabled under `xy_periodic` (the periodic images have no finite-distance observer); the emergent observable is the boundary intensity  $I(\mu)$  (§3.3).

| parameter                        | default | meaning                                                                                                                                                                                                                |
|----------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| slab_faces                       | 'top'   | which z-face(s) are lit: 'top', 'bottom', or 'both'                                                                                                                                                                    |
| slab_top_mode, slab_bot_mode     | 'beam'  | per face: 'beam' (collimated) or 'isotropic' (Lambert)                                                                                                                                                                 |
| slab_top_theta, slab_bot_theta   | 0       | beam incidence angle from the inward normal [deg]; 0 is normal incidence                                                                                                                                               |
| slab_top_phi, slab_bot_phi       | 0       | beam azimuth [deg]                                                                                                                                                                                                     |
| slab_top_source, slab_bot_source | 0       | per-face source strength (band-integrated): the specific intensity $I$ [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] for 'isotropic', or the beam flux $F_n$ [ $\text{erg s}^{-1} \text{cm}^{-2}$ ] for 'beam' |
| slab_nmu                         | 20      | number of $\mu$ -bins of the emergent $I(\mu)$ output                                                                                                                                                                  |

The spectrum is the global one (tstar / ion\_spectrum). Both z-faces always escape (no reflecting boundary). launch\_sequence = 'sobol' is supported (both modes). With dust scattering (ion\_dust\_scatter) the scattered/reflected component is included in the boundary  $I(\mu)$ .

## 2.5 Spectrum-file formats

A single parameter, spectrum\_type (default 'shape'), fixes the column units of *every* spectrum-file slot (ion\_spectrum, src\_spectrum\_file, ext\_spectrum). The type sets the units; the slot sets the meaning: an internal-source file gives a luminosity density  $L_X$ , the external-field file gives an intensity density  $J_X$ . Column 1 and the value columns are:

| spectrum_type | column 1                    | internal (cols 2+)                                   | external (col 2) |
|---------------|-----------------------------|------------------------------------------------------|------------------|
| 'shape'       | $E$ [eV] $\uparrow$         | arbitrary per eV (shape of $L_E$ )                   | arbitrary per eV |
| 'per_ev'      | $E$ [eV]                    | $L_E$ [ $\text{erg s}^{-1} \text{eV}^{-1}$ ]         | $J_E$            |
| 'per_hz'      | $\nu$ [Hz]                  | $L_\nu$ [ $\text{erg s}^{-1} \text{Hz}^{-1}$ ]       | $J_\nu$          |
| 'per_ang'     | $\lambda$ [ $\text{\AA}$ ]  | $L_\lambda$ [ $\text{erg s}^{-1} \text{\AA}^{-1}$ ]  | $J_\lambda$      |
| 'per_um'      | $\lambda$ [ $\mu\text{m}$ ] | $L_\lambda$ [ $\text{erg s}^{-1} \mu\text{m}^{-1}$ ] | $J_\lambda$      |

The external intensity density  $J_X$  carries the same spectral unit as the internal  $L_X$  times an extra  $\text{cm}^{-2} \text{sr}^{-1}$  (a mean intensity). All files are plain text with '#' comment lines; values outside the tabulated range are zero. src\_spectrum\_file is one multi-column file: column 1 shared by all sources, then one value column for each source (extra columns beyond nsource are ignored with a note, too few abort).

**The abscissa is a sample grid, not bin centers.** Column 1 lists the points at which the spectral density is *sampled*: the file describes a continuous function, defined between the points by linear interpolation, and no bin widths are attached to the rows. MoCHII integrates each of its own band bins on a 32-point sub-grid of the bin, interpolating the table at every sub-point (zero outside the tabulated range), so the file grid and the band grid are independent. The spacing is arbitrary — nonuniform grids are fine; place points densely where the spectrum varies rapidly, since only linear variation is captured between adjacent points. Abscissa values must be distinct (no duplicates); a 'shape' file must be ascending in  $E$ , while

the physical types accept any row order (the table is sorted internally after the unit conversion, so a wavelength table may be given wavelength-ascending).

**Absolute vs. shape normalization.** A physical type ('per\_ev'/'per\_hz'/'per\_ang'/'per\_um') is *absolute*: the bin luminosities follow the tabulated integral directly. If the matching scale (`luminosity / src_lum(i) / ext_intensity`) is unset it is DERIVED from the file — each source from its column's ionizing-segment integral, the external field from its band-integrated  $J$  — and reported in the log. If a scale is set, the spectrum is RESCALED to it (a note is printed) and the file then acts only as a shape. With `add_fuv` an absolute spectrum fills the FUV bins from the same table, with no separate normalization. 'shape' keeps the legacy behavior: the shape is normalized to the scale in every case. A spectrum with no luminosity inside the band aborts, as does rescaling a file whose ionizing-segment integral is zero.

**'shape' is per eV, not variable-agnostic.** 'shape' reads its value columns as the shape of the per-eV density  $L_E$  on the  $E$  [eV] grid of column 1. A table of  $F_\lambda$  (or  $F_\nu$ ) values re-gridded onto  $E$  under 'shape' therefore omits the Jacobian and is wrong by a factor  $E^{-2}$  (the  $d\lambda/dE$  conversion). To use a wavelength- or frequency-tabulated shape, give it with its matching physical type and set the scale explicitly — the type conversion then applies the Jacobian and the scale RESCALES the file to your target, so the file acts as a shape with the correct units. For example, a wavelength shape normalized to  $10^{49}$  ionizing photons' worth of luminosity:

```
par%spectrum_type = 'per_ang'
par%ion_spectrum   = 'wave_shape.txt'
par%luminosity     = 3.0e38
```

A small 'per\_ang' source file for two point sources ( $\lambda$  ascending; the second source is FUV-faint):

```
# lambda[A]   L_lam,1   L_lam,2
300.0         7.80e34   3.4e33
500.0         6.05e34   1.2e33
912.0         4.10e34   0.0
```

## 2.6 Initial state and iteration

| parameter             | default | meaning                                                                                                                                                  |
|-----------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| xHI_init              | 1.0     | initial $n_{\text{HI}}/n_{\text{H}}$ (an xHI file column wins); an ionized start (1e-3) converges much faster than neutral                               |
| xHeI_init, xHeII_init | 1, 0    | initial He fractions                                                                                                                                     |
| He_abund              | 0.1     | $n_{\text{He}}/n_{\text{H}}$                                                                                                                             |
| gas_niter             | 0       | iterations (0 = a single rates-only pass)                                                                                                                |
| gas_tol               | 1e-3    | convergence tolerance on $x_{\text{HII}}$ (measure set by <code>conv_crit</code> )                                                                       |
| conv_crit             | 'cell'  | convergence measure: 'cell' (cell maxima; stalls at the front-cell noise floor) or 'vol' (volume-integrated $L_1$ ; noise-robust, recommended). See §2.7 |
| ion_relax             | 1.0     | under-relaxation on $x$ ( $< 1$ damps front oscillation)                                                                                                 |



| parameter           | default | meaning                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| case_ab             | 'B'     | case A/B recombination in the ionization balance, i.e. how the ionizing <i>continuum</i> is treated ('A' is forced by diffuse_field, which transports the ground-state recombination photons)                                                                                                                                                                                                                            |
| line_case           | 'B'     | case of the SH95 H I/He II recombination- <i>line</i> tables. This is set by the optical depth of the Lyman <i>lines</i> , a different question from case_ab: a nebula whose diffuse continuum is transported still traps its Lyman lines and radiates case-B Balmer lines, so 'B' is the physical default for H II regions. 'A' selects the case-A tables (Lyman-line-thin gas); 'follow' ties the line case to case_ab |
| require_convergence | F       | if the gas iteration has not converged at gas_niter, print a warning; when T, abort before writing output. The convergence status is recorded in the _rates header either way (CONVERGD, NITERDON, FINALDX, FINALDTE)                                                                                                                                                                                                    |

## 2.7 Convergence of the gas iteration

The gas iteration (gas\_niter sweeps of transport → rate integrals → leaf solve → opacity refill) stops when the state stops changing between sweeps. Each sweep compares a *change measure* against a tolerance:

- ionization only (te\_fixed, no thermal solve): converged when the  $x_{\text{HII}}$  change  $< \text{gas\_tol}$ ;
- coupled (solve\_te): converged when the  $x_{\text{HII}}$  change  $< \text{gas\_tol}$  *and* the  $T_e$  change  $< \text{gas\_tol\_te}$ .

**Two measures (conv\_crit).** Both are computed every sweep; conv\_crit selects which one gates the loop.

**'cell'** the cell maxima  $\max_{\text{leaf}} |\Delta x_{\text{HII}}|$  and  $\max_{\text{leaf}} |\Delta T_e|/T_e$ . Simple, but the ionization-front cells jitter with Monte Carlo noise and never settle, so  $\max |\Delta x_{\text{HII}}|$  *stalls* at the front-cell noise floor ( $\sim 2\text{--}4 \times 10^{-2}$  at  $4 \times 10^7$  packets) and a smaller gas\_tol is unreachable. This is the historical default (kept so the recorded gates reproduce).

**'vol'** the volume-integrated  $L_1$  changes

$$\frac{\sum_{\text{leaf}} V |\Delta x_{\text{HII}}|}{\sum_{\text{leaf}} V x_{\text{HII}}} < \text{gas\_tol}, \quad \frac{\sum_{\text{leaf}} V n_e |\Delta T_e|}{\sum_{\text{leaf}} V n_e T_e} < \text{gas\_tol\_te}.$$

Front jitter occupies little volume (a few thousand front cells against  $\sim 10^5$  ionized ones), so it enters the norm suppressed by that ratio; the  $n_e$  weight drops the vacuum and te\_min-pinned no-heating cells that otherwise dominate  $\max |\Delta T_e|$  in FUV/PDR runs. These measures decay smoothly — the recommended production setting.

**The Monte Carlo floor.** Even the volume measures do not reach zero: they floor at a Monte Carlo level set by the packet count, scaling as  $1/\sqrt{N_{\text{packets}}}$  ( $\sim 3 \times 10^{-3}$  in  $x$  and  $\sim 7 \times 10^{-3}$  in  $T_e$  at  $8 \times 10^6$  packets). Reaching the floor is convergence — iterating further leaves the solution unchanged. So **set gas\_tol/gas\_tol\_te just above the floor for**

**the packet count in use:** a tolerance below the floor can never be met and the loop runs to `gas_niter` and warns. Rules of thumb (`conv_crit='vol'`):  $\sim 5 \times 10^{-3}/10^{-2}$  at  $\sim 10^7$  packets, loosened to  $\sim 10^{-2}/2 \times 10^{-2}$  at  $\sim 10^6$  (the reduced-count examples/ use the looser values for this reason).

**When it does not converge.** Hitting `gas_niter` without meeting the test is not fatal: the state is still written and a warning is printed on rank 0. Set `require_convergence = T` to abort before the output instead. Either way the `_rates` header records the outcome in `CONVERGD` (0/1), `NITERDON` (sweeps run), `FINALDX`, and `FINALDTE`. The derivation and the gate measurements are in `docs/MoCHII_physics.pdf`.

## 2.8 Thermal balance and metals

| parameter                     | default       | meaning                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>solve_te</code>         | F             | bisection on $T_e$ coupled with the ionization                                                                                                                                                                                                                                                                                                                   |
| <code>te_fixed</code>         | 1e4           | fixed $T_e$ [K] when <code>solve_te</code> is off                                                                                                                                                                                                                                                                                                                |
| <code>te_min, te_max</code>   | 1e3, 5e4      | bisection bracket [K]                                                                                                                                                                                                                                                                                                                                            |
| <code>gas_tol_te</code>       | 1e-3          | convergence on $\max  \Delta T_e /T_e$                                                                                                                                                                                                                                                                                                                           |
| <code>cooling_model</code>    | 'local_ne'    | line cooling in the thermal loop: 'local_ne' (default) suppresses each ion's cooling fit to the local $n_e$ with a $(T, n_e)$ table built from its $n$ -level solve, so the thermal balance and the emergent lines are one physics; 'low_density' uses the $n_e \rightarrow 0$ fits directly (the coronal limit, exact in diffuse gas; reproduces earlier gates) |
| <code>atomic_dir</code>       | 'data/atomic' | coefficient files (relative to the run directory)                                                                                                                                                                                                                                                                                                                |
| <code>use_metals</code>       | F             | registry cascade + metal line cooling                                                                                                                                                                                                                                                                                                                            |
| <code>ci_model</code>         | 'voronov'     | collisional ionization: 'voronov' (default) or 'dere_hybrid' (Voronov $\times$ the Dere 2007/Voronov ratio; matters only in shielded low-ionization zones)                                                                                                                                                                                                       |
| <code>abund_C/N/O/Ne/S</code> | HII20 values  | $n(X)/n(H)$ ; an element is active when $> 0$                                                                                                                                                                                                                                                                                                                    |
| <code>abund_Ar/Mg/Fe</code>   | 0             | further registry elements (gas-phase values; Mg/Fe are heavily depleted)                                                                                                                                                                                                                                                                                         |
| <code>abund_Si/Cl/Ca</code>   | 0             | five-stage registry elements (gas-phase; Si and especially Ca heavily depleted, Cl nearly undepleted — Orion-like $3e-6 / 3e-7 / 2e-8$ ); adds [Si II] 34.8 $\mu m$ , Si III/IV, [Cl II/III/IV] (5518/5538 density pair), Ca II, [Ca V]                                                                                                                          |
| <code>ion_metal_abs</code>    | T             | metal photoionization absorption in the band opacity (active with <code>use_metals</code> ); the only gas opacity in the FUV bins                                                                                                                                                                                                                                |

| parameter       | default   | meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| charge_exchange | 'two_way' | charge exchange $X^{i+} + H^+ \leftrightarrow X^{(i+1)+} + H^0$ in hydrogen's balance as well as the cascade's, so the charge each reaction moves is conserved in the reacting pair: $n_{\text{HI}}(\Gamma_{\text{H}} + n_e C_{\text{H}} + R_{\text{cx}}^{\text{out}}) = (\alpha_{\text{H}} n_e + R_{\text{cx}}^{\text{in}}) n_{\text{HII}}$ with the sums over all 29 active reactions. 'metal_only' keeps them in the cascade alone (the earlier bookkeeping, which does not conserve charge); it is a reproduction value, not a physics option |

| parameter            | default | meaning                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| emis_output          | F       | write _emis (HDF5/FITS): leaf-by-leaf line emissivities + the state needed for maps (Section 3.)                                                                                                                                                                                                                                                                      |
| hei_metastable       | F       | write _heimeta (HDF5/FITS): the He I $2^3\text{S}$ metastable density $n_3$ , its photoionization rate $\Phi_3$ , plus $n_{\text{H}}$ , $n_e$ , $T_e$ , and leaf geometry (the 10833 Å absorption diagnostic; Section 3.). With add_fuv and efuv_min=4.78 the soft-UV part of $\Phi_3$ is included                                                                    |
| h_coll_effects       | F       | add the collisional H I Balmer component (excitation of neutral H) as separate hc rows/blocks; matters at fronts and hot cells                                                                                                                                                                                                                                        |
| fe_levels_full       | F       | load the expanded Fe II/III models (nlevel_fe_{2,3}_full.txt, 16/34 levels) instead of the compact 13/14 for iron-line studies                                                                                                                                                                                                                                        |
| diffuse_field        | F       | explicit ground-recombination packets                                                                                                                                                                                                                                                                                                                                 |
| diffuse_energy_model | 'milne' | spectrum of the three ground continua (needs diffuse_field): 'milne' (default) is the detailed-balance free-bound spectrum $E^2 \sigma_{\text{pi}}(E) e^{-(E-E_{\text{th}})/kT}$ with the transport cross sections; 'exponential' the historical $E_{\text{th}} + kT[-\ln u]$ approximation (reproduces earlier runs); 'threshold' a sensitivity bracket, not a model |

| parameter      | default | meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hei_diffuse    | F       | fourth diffuse channel (needs diffuse_field): He I excited-level recombination photons. The $2^3S:2^1S:2^1P$ branch fractions come from one consistent set of $\alpha_{\text{eff}}$ (Benjamin, Skillman & Smits 1999 Table 1 with J. S. Mathis, as in K. Wood’s ionize) normalized by its own sum — $0.762:0.076:0.162$ at 8000 K against the statistical $0.750:0.083:0.167$ — and the $2^1S$ photons from the Drake et al. (1969) two-photon shape. Correct energy bookkeeping: +0.12% in energy and +0.30% in photons per case-B He II recombination, $L(H\beta)$ within 0.02%, He split unchanged. The trapped $2^1P$ (584 Å) photon is split locally between photoionizing hydrogen and collisional conversion $2^1P \rightarrow 2^1S$ , with $p_{584} = [1 + 77x(\text{He}^0)/(\sqrt{T}x(\text{H}^0))]^{-1}$ (Wood, Mathis & Ercolano 2004) — the same split the He I two-photon nebular continuum uses. At HII40 it lowers $L(H\beta)$ by 0.2% and moves the He I front inward by 0.0036 pc |
| gaunt_vh14     | F       | free–free Gaunt factors from the van Hoof et al. (2014) table (data/gauntff_vh14.dat) instead of Hummer (1988); the two agree to $\lesssim 0.5\%$ in $T_e$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| metal_ne       | F       | PDR: electrons from the metal cascade join the $n_e$ closure (the only electrons beyond the I-front)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| metal_heat     | F       | PDR: metal photoheating in the thermal balance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| grain_pe       | F       | PDR thermal package: grain photoelectric heating + recombination cooling (Bakes & Tielens 1994) from the local FUV field (needs add_fuv), plus H-impact [C II] $158\text{ }\mu\text{m}$ / [O I] $63\text{ }\mu\text{m}$ cooling; scaled by pe_scale and the leaf dust amount                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| pe_scale       | 1.0     | photoelectric-efficiency scale factor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| use_sec_ion    | F       | secondary ionization by fast photoelectrons (Shull & van Steenberg 1985): a photoelectron with kinetic energy $E_0 = h\nu - E_{\text{th}}$ above 40 eV deposits only $f_{\text{heat}}(x)$ of its excess as heat and the balance drives secondary H I/He I ionizations ( $x$ = ionized fraction of the H+He nuclei); $\sim 0$ in fully ionized gas, matters at fronts / PDR / hard fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| metal_freefree | T       | include the trace metals in the free–free cooling with the net ion charge $Z_{\text{ion}} = (\text{stage} - 1)$ , not the nuclear $Z$ ; a trace ( $\sim 10^{-4}$ of the H/He free–free) correctness term, on by default                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 2.9 I-front re-refinement

| parameter                 | default | meaning                                             |
|---------------------------|---------|-----------------------------------------------------|
| refine_front              | F       | rebuild the octree on the I-front once              |
| refine_iter               | 20      | at which iteration                                  |
| refine_lbase, refine_lmax | 5, 7    | base/front levels                                   |
| refine_eps                | 1e-2    | front flag: $\epsilon < x_{\text{HI}} < 1-\epsilon$ |
| refine_dx                 | 0.2     | or a face-neighbor $x_{\text{HI}}$ jump above this  |

## 2.10 Dust in the band and dust emission

| parameter                | default          | meaning                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ion_add_dust             | F                | dust absorption in the band (dusty Strömgren)                                                                                                                                                                                                                                                                                                  |
| ion_dust_kext            | "                | kext table with EUV coverage (e.g. data/kext_albedo_WD_MW_3.1_60_D03.all_2003); required                                                                                                                                                                                                                                                       |
| ion_dust_scatter         | F                | sample dust scattering (HG, albedo weights)                                                                                                                                                                                                                                                                                                    |
| dust_sed                 | F                | SEDust stochastic dust SED at output time                                                                                                                                                                                                                                                                                                      |
| dust_model_sed           | 'astrodust'      | 'astrodust'/'dl07'/'zubko'                                                                                                                                                                                                                                                                                                                     |
| sed_workdir              | " (auto)         | SEDust sed/ tree whose dielectric tables are read relative to it; blank auto-resolves to <executable dir>/SEDust/sed (the in-tree copy), so it normally needs no setting                                                                                                                                                                       |
| sed_qtable, sed_sizedist | in-tree defaults | optics/size-distribution tables (relative to sed_workdir)                                                                                                                                                                                                                                                                                      |
| sed_NT, sed_Tlo, sed_Thi | 200, 2.7, 5e3    | SEDust temperature grid                                                                                                                                                                                                                                                                                                                        |
| dust_emis_bands          | unset            | up to 8 wavelengths [ $\mu\text{m}$ ] (e.g. 8, 24, 70, 160): SEDust band emissivities of each leaf $\rightarrow$ _dustemis (blocks emis_dust/wl_dust [ $\text{erg s}^{-1} \text{cm}^{-3} \mu\text{m}^{-1}$ ]); the Python map maker projects them into dust-band images (the IR is optically thin)                                             |
| sed_pah_live             | F                | $x_{\text{HII}}$ -dependent PAH survival in the emission: the leaf spectrum is assembled from the model channels with the PAH channel weighted by $(x_{\text{HI}} + f_{\text{ion,pah}} x_{\text{HII}})$ (needs a model with a 'PAH' channel, i.e. astrodust); turns the 8- $\mu\text{m}$ map into the classic PAH ring at the ionization front |

## 2.11 Peel-off imaging

| parameter        | default  | meaning                                                                                                                                                                                                                                                                                                                   |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ion_peel         | F        | after convergence, one extra transport pass peels direct (stellar + diffuse) and dust-scattered contributions toward the observers $\rightarrow$ <code>_image</code> (blocks <code>direct_ion/scatt_ion</code> , plus <code>_fuv</code> with <code>add_fuv</code> ) [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] |
| peel_bins        | F        | bin-resolved cubes <code>direct_cube/scatt_cube</code> ( $n_x, n_y, n_z$ ) in addition to the band-integrated channel images (hardness maps, synthetic EUV/FUV photometry); the third (bin) axis is labeled by the <code>E_bin/dE_bin</code> [eV] and <code>wl_bin</code> [Å] arrays written once into the same file      |
| save_direct0     | F        | also write the unattenuated direct image ( <code>direct0_ion/_fuv</code> , and <code>direct0_cube</code> with <code>peel_bins</code> ): the direct source with the path extinction $e^{-\tau}$ omitted, so <code>direct0/direct</code> = $e^{+\tau}$ is the line-of-sight optical depth toward the observer               |
| obsx, obsy, obsz | unset    | observer positions (or directions) in code units; or use <code>alpha/beta/gamma</code> angle lists [deg]; default: one observer on +z far away                                                                                                                                                                            |
| distance         | auto     | observer distance (code units)                                                                                                                                                                                                                                                                                            |
| nxim, nyim       | 129, 129 | image pixels                                                                                                                                                                                                                                                                                                              |
| dxim, dyim       | auto     | pixel scale [deg]; auto-sized to cover the box                                                                                                                                                                                                                                                                            |

Peel-off also works with the external field: an external packet's direct peel uses the Lambert weight  $\max(\hat{k}_{\text{obs}} \cdot \hat{n}_{\text{in}}, 0)/\pi$  about its entry-surface inward normal, so only the far-side surface contributes and the image is the background field seen through the medium (the cloud in silhouette); with `save_direct0`, `direct0/direct` =  $e^{+\tau}$  is again the line-of-sight optical-depth map.

## 2.12 Output

`out_file` (base name; extension follows `file_format` = 'hdf5' or 'fits').

## 3. Output files

### 3.1 <base>\_rates.h5

One dataset for each extension, leaf-ordered:

| extension                     | content                                                                  |
|-------------------------------|--------------------------------------------------------------------------|
| Gamma_HI/HeI/HeII             | photoionization rates [ $\text{s}^{-1}$ ]                                |
| Heat_HI/HeI/HeII              | photoheating per particle [ $\text{erg/s}$ ]                             |
| Heat_dust                     | grain heating [ $\text{erg s}^{-1} \text{cm}^{-3}$ ] (when ion_add_dust) |
| T_dust                        | equilibrium mixture dust temperature [K]                                 |
| x_HI, x_HeI, x_HeII, n_e, T_e | converged state                                                          |
| x_<el>_stages                 | metal ion fractions (stage, leaf)                                        |
| J_nu                          | band mean intensity (bin, leaf) [ $\text{erg/s/cm}^2/\text{Hz/sr}$ ]     |
| E_bin, dE_bin                 | band grid [eV]                                                           |
| LeafXYZ, LeafSize             | leaf centers and full widths (code units; for slices/cutouts)            |

The file header also carries the convergence record of the gas iteration: CONVERGD (did it converge), NITERDON (iterations run), FINALDX (final max  $|\Delta x_{\text{HII}}|$ ), and FINALDTE (final max  $|\Delta T_e|/T_e$ ).

### 3.2 Line and emissivity outputs

<base>\_lines.txt is a text table of integrated line luminosities on the converged state, one row for each line: element, ionization stage, wavelength [ $\text{\AA}$ ],  $L$  [ $\text{erg s}^{-1}$ ], and  $L/L(\text{H}\beta)$ . It is written by every run: the H and He recombination lines need only  $n_e$ ,  $T_e$  and the He ionization state, so an H/He model at fixed temperature gets its table too, with the metal blocks simply absent. With emis\_output the same line set is also written leaf-by-leaf to <base>\_emis (HDF5, or FITS with file\_format='fits') as emissivity blocks in  $\text{erg s}^{-1} \text{cm}^{-3}$ , where  $\sum \text{emis} \cdot V = L_{\text{line}}$  and the block row order matches the text table. Every line comes from one of two sources: recombination-line tables (H and He, Table 13) and  $n$ -level statistical-equilibrium solves (the metals plus the optional collisional H I component, Table 14).

Table 13: Recombination-line blocks in <base>\_emis; wavelengths in  $\text{\AA}$  unless  $\mu\text{m}$  noted.

| block     | source                                                                           | lines                                                                                                                                                                 | weighting,<br>present                                    | when<br>present |
|-----------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------|
| emis_h    | Storey & Hummer (1995) H I, case set by line_case                                | H $\alpha$ 6565, H $\beta$ 4863, H $\gamma$ 4342, H $\delta$ 4103, Pa $\alpha$ 1.876 $\mu\text{m}$ , Pa $\beta$ 1.282 $\mu\text{m}$ , Br $\gamma$ 2.166 $\mu\text{m}$ | always; $n_e n_{\text{HII}}$                             |                 |
| emis_heii | SH95 He II (case per line_case)                                                  | 1640, 3204, 4543, 4687, 5413, 10126                                                                                                                                   | only where an He III zone exists; $n_e n_{\text{HeIII}}$ |                 |
| emis_hei  | Porter et al. (2012/2013) He I, case-B collisional-radiative (density-dependent) | 3890, 4027, 4473, 5877, 6680, 7067, 7283, 10833                                                                                                                       | where He II is present; $n_e n_{\text{HeII}}$            |                 |
| emis_hc   | 25-level H I collisional solve                                                   | H $\alpha$ /H $\beta$ /H $\gamma$ collisional component (separate hc rows)                                                                                            | only h_coll_effects; $n_{\text{HI}}$                     | with            |

The He I Porter table is a case-B collisional-radiative total (density-dependent; it already includes collisional excitation from the  $2^3\text{S}$  metastable), case B only; under case A a note is printed and the H I/He II tables switch to case A while He I stays case B.

The metal line blocks `emis_<el>_<stage>` follow the  $n$ -level solve. One block is written for every active registry ion (element abundance  $> 0$ ) that has an `nlevel_<el>_<stage>` data file. Elements C, N, O, Ne, S are active by default; Ar, Mg, Fe, Si, Cl, Ca are activated by setting their `par%abund_<El>`. Only the principal lines are listed below — the full transition list of each block is in `<base>_lines.txt`.

Table 14: Metal collisional-line blocks (`emis_<el>_<stage>`); wavelengths in Å unless  $\mu\text{m}$  noted.

| ion (block)                                                       | principal lines                                                                                       |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| c_2 [C II]                                                        | 2325 (UV multiplet), 158 $\mu\text{m}$                                                                |
| c_3 [C III]                                                       | 977, 1907, 1909, 178 $\mu\text{m}$                                                                    |
| n_2 [N II]                                                        | 3064/3071, 5756, 6550, 6585, 122 $\mu\text{m}$ , 205 $\mu\text{m}$                                    |
| n_3 [N III]                                                       | 1750 (UV multiplet), 57 $\mu\text{m}$                                                                 |
| o_1 [O I]                                                         | 5579, 6302, 6366, 63 $\mu\text{m}$ , 145 $\mu\text{m}$                                                |
| o_2 [O II]                                                        | 2471, 3727, 3730, 7321/7331                                                                           |
| o_3 [O III]                                                       | 1661/1666, 2322/2332, 4364, 4960, 5008, 52 $\mu\text{m}$ , 88 $\mu\text{m}$                           |
| ne_2 [Ne II]                                                      | 12.8 $\mu\text{m}$                                                                                    |
| ne_3 [Ne III]                                                     | 3343, 3870, 3969, 15.55 $\mu\text{m}$ , 36 $\mu\text{m}$                                              |
| s_2 [S II]                                                        | 4070/4078, 6718, 6733, 1.03 $\mu\text{m}$ , 315 $\mu\text{m}$                                         |
| s_3 [S III]                                                       | 3722, 6314, 8831, 9070, 9532, 12 $\mu\text{m}$ , 18.7 $\mu\text{m}$ , 33.5 $\mu\text{m}$              |
| <i>Optional — activate with <code>par%abund_&lt;El&gt;</code></i> |                                                                                                       |
| ar_3 [Ar III]                                                     | 3006/3110, 5193, 7138, 7753, 8.99 $\mu\text{m}$ , 21.8 $\mu\text{m}$                                  |
| ar_4 [Ar IV]                                                      | 2854/2869, 4713, 4741, 7170 (multiplet), 56 $\mu\text{m}$ , 78 $\mu\text{m}$                          |
| mg_2 [Mg II]                                                      | 2796, 2804                                                                                            |
| fe_2 [Fe II]                                                      | 1.26/1.32 $\mu\text{m}$ , 5.34 $\mu\text{m}$ , 17.9 $\mu\text{m}$ , 26 $\mu\text{m}$ (40 lines total) |
| fe_3 [Fe III]                                                     | 4659, 4703, 4881, 5011, 22.9 $\mu\text{m}$ (38 lines; 16/34-level with <code>fe_levels_full</code> )  |
| si_2 [Si II]                                                      | 2335 (multiplet), 34.8 $\mu\text{m}$                                                                  |
| si_3 [Si III]                                                     | 1206, 1883, 1892, 38 $\mu\text{m}$                                                                    |
| si_4 [Si IV]                                                      | 1394, 1403                                                                                            |
| cl_2 [Cl II]                                                      | 3587/3679, 6164, 8581, 9126, 14.4 $\mu\text{m}$ , 33 $\mu\text{m}$                                    |
| cl_3 [Cl III]                                                     | 3344/3354, 5519, 5539, 8480 (multiplet)                                                               |
| cl_4 [Cl IV]                                                      | 3120/3205, 5325, 7263, 7533, 11.8 $\mu\text{m}$ , 20.3 $\mu\text{m}$                                  |
| ca_2 [Ca II]                                                      | 3935, 3970, 7293/7326, 8500/8544/8665                                                                 |
| ca_5 [Ca V]                                                       | 3999, 5311, 6088, 3.05 $\mu\text{m}$ , 4.16 $\mu\text{m}$ , 11.5 $\mu\text{m}$                        |

Some registry ions are tracked in the ionization balance but carry no line block because CHIANTI v11 has no level/collision data for them: Ar II, Cl V, Ca III, Ca IV.



**State blocks.** Beyond the emissivities, <base>\_emis carries the state a map needs without the rates file: LeafXYZ (leaf centers) with the DIST\_CM keyword, LeafSize (leaf edge length;  $V = (\text{LeafSize} - \text{DIST\_CM})^3$ ), n\_H, n\_e, T\_e, x\_HI, x\_HeI, x\_HeII, and the x\_<el>\_stages metal-fraction blocks (same layout as the rates file). The Python EmisData reader (tools/python/mochii\_output.py) turns these blocks into line maps without the rates file.

### 3.3 Spectral and image outputs

| file          | content                                                                                                                                                                                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _nebcont.txt  | nebular continuum $L_\nu$ (fb+ff tables, Milne, two-photon)                                                                                                                                                                                                                                                                                                    |
| _dustir.txt   | equilibrium-mixture dust IR spectrum (1–1000 $\mu\text{m}$ ) with the energy-closure header                                                                                                                                                                                                                                                                    |
| _dustsed.txt  | SEDust stochastic dust SED with PAH bands (when dust_sed)                                                                                                                                                                                                                                                                                                      |
| _dustemis.h5  | (with dust_emis_bands) SEDust band emissivities of each leaf, same block layout as _emis — readable by EmisData for dust-band maps                                                                                                                                                                                                                             |
| _heimeta.h5   | (with hei_metastable) the He I $2^3\text{S}$ diagnostic: n_2s3 [ $\text{cm}^{-3}$ ] (with the DIST_CM and A31 keywords), Phi_3 [ $\text{s}^{-1}$ ], n_H, n_e, T_e, LeafXYZ, LeafSize — read by the Python HeiMetaData class for column and 10833 Å optical-depth maps, and convertible to a LaRT scattering-medium input by heimeta_to_lart.py                 |
| _image.h5     | (with ion_peel) peel-off images for each observer: direct_ion/scatt_ion (+_fuv) [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ], suffix _obs<k> when several observers; with peel_bins, direct_cube/scatt_cube ( $n_x, n_y, n_\nu$ ) and the E_bin/dE_bin/wl_bin bin-grid arrays; with save_direct0, the unattenuated direct0_ion/_fuv (or direct0_cube) |
| _slab_Imu.txt | (with source_geometry = 'slab') the emergent intensity $I(\mu)$ at the two z-boundaries: columns $\mu$ , $I_{\text{top}}$ , $I_{\text{bot}}$ [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] on slab_nmu bins ( $F = 2\pi \int I \mu d\mu$ ), with the reflected / transmitted / absorbed energy budget (fractions of the entering $L$ ) in the header   |

### 3.4 Reading outputs in Python

tools/python/mochii\_output.py is an importable module (works directly in a notebook) that reads every section file (HDF5 or FITS; read\_sections), the line table (read\_lines\_table), and the text spectra (read\_spectrum), and builds 2D maps from the emissivity file:

```
import sys; sys.path.insert(0, ".../MoCHII/tools/python")
import matplotlib.pyplot as plt
from mochii_output import EmisData

d = EmisData("run_emis.h5")
d.info()                                # blocks, fields, line count
```

```

d.line_list()                                # every stored line (block, wl)

# ready-made figures (notebook): returns (fig, ax)
fig, ax = d.plot_line("o_3", 5007.)          # log S map
fig, ax = d.plot_line("h", 6563., unit="intensity",
                      photons=True)         # photons/s/cm^2/sr
fig, ax = d.plot_field("T_e", weight="EM")
fig, axs = plt.subplots(1, 2); d.plot_line("s_2", 6717., ax=axs[0]) \
    ; d.plot_line("s_2", 6731., ax=axs[1])   # doublet side by side

# raw arrays, when the figure is yours to build
img, ext = d.line_map("o_3", 5007., axis="z", npix=512)
img, ext = d.line_map("h", 6563., unit="intensity", photons=True)
img, ext = d.field_map("T_e", weight="EM")   # LOS-weighted average
Q = d.line_luminosity("h", 4861., photons=True) # photon rate [1/s]

```

Line maps are flux-conserving: each leaf's luminosity is deposited with exact area overlap onto the pixel grid (AMR leaf sizes handled), so  $\sum S A_{\text{pix}} = L_{\text{line}}$  to machine precision. `unit='flux'` (default) gives surface brightness [ $\text{erg s}^{-1} \text{cm}^{-2}$ ]; `unit='intensity'` gives specific intensity [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] =  $S/4\pi$  (isotropic emission, optically thin); `photons=True` converts either to photon units (divides by  $hc/\lambda$  of the line). A `slab=(lo,hi)` cut restricts the line of sight. Field maps average `T_e`, `n_e`, `x_HI`, ... along the line of sight with weight 'EM' ( $n_e^2 V$ ), 'ne', 'nH', or 'V'. The same module doubles as a command-line tool for quick looks:

```

python3 mochii_output.py run_emis.h5          # list lines
python3 mochii_output.py run_emis.h5 --line o_3 5007 \
    --npix 512 --log --out o3.png
python3 mochii_output.py run_emis.h5 --line h 6563 \
    --unit intensity --photons                # photons/s/cm^2/sr
python3 mochii_output.py run_emis.h5 --field T_e --weight EM

```

### 3.5 Slices and cutouts

A *slice* is the actual leaf value on a plane (the cell that contains each pixel), distinct from the line-of-sight *projection* above — a slice resolves the ionization front sharply where a projection blurs it. `tools/python/leaf_field.py` holds the backend (`leaf_slice`, the nearest-leaf fallback `leaf_slice_nn`, and `plot_slice`); `mochii_output.py` exposes it on the output files:

```

from mochii_output import EmisData, RatesData
r = RatesData("run_rates.h5")          # leaf state
fig, ax = r.plot_field_slice("T_e", axis="z", coord=0.0)
fig, ax = r.plot_field_slice("x_HI", axis="z", coord=0.0, log=True)

```

```
img, ext = r.slice_field("n_e", axis="x", coord=0.0,
                        bounds=(-2,2,-2,2)) # cutout window
```

```
d = EmisData("run_emis.h5")
fig, ax = d.plot_line_slice("o_3", 5007., axis="z") # emissivity slice
```

Exact slices need the leaf full widths: the `_emis.h5` file always carries `LeafSize`, and `_rates.h5` does too; a rates file written before `LeafSize` existed falls back to a nearest-leaf slice (a one-time note is printed). The command line takes `--slice FIELD --axis z --coord 0` on either file.

### 3.6 He I 2<sup>3</sup>S metastable and 10833 Å absorption

The `_heimeta` file (`hei_metastable`) is read by the `HeiMetaData` class, which adds a line-of-sight column map and the 10833 Å line-center optical depth:

```
from mochii_output import HeiMetaData
d = HeiMetaData("run_heimeta.h5")
d.info() # n_2s3, Phi_3, line constants
img, ext = d.column_map("n_2s3") # LOS column [cm^-2]
img, ext = d.tau10833_map(v_turb=0.0, # tau_0 map (km/s turbulence)
                        component="doublet") # blended J'=1,2 pair
fig, ax = d.plot_column("n_2s3", log=True)
fig, ax = d.plot_tau10833(component="all")
```

`column_map` is the same flux-conserving deposit as the line maps with weight  $nV$ , so  $\sum NA_{\text{pix}} = \sum nV$  to machine precision. `tau10833_map` deposits the line-center opacity  $\kappa_0 = \frac{\sqrt{\pi} e^2}{m_e c} f \frac{\lambda}{b} n_3$ ,  $b = \sqrt{2kT_e/m_{\text{He}} + v_{\text{turb}}^2}$ , so a pixel value is  $\int \kappa_0 dl$ . `component` selects the resolved  $J' = 0$  line (10832.06 Å), the blended  $J' = 1,2$  'doublet' (10833.22/10833.31 Å, 0.089 Å apart, inside the thermal Doppler width), or 'all'; the constants ( $\lambda, f, A$ ) come from `data/atomic/hei_metastable.txt`. The map warns when line-center  $\tau_0 > 1$ : the diagnostic assumes optically thin 10833 (line transfer is out of scope). The command line takes `--column n_2s3` or `--tau10833 --component doublet --vturb 5`.

Where line transfer *is* wanted (resonant-scattering emission profiles from the metastable medium), `heimeta_to_lart.py` (in `tools/python/`) converts a `_heimeta` file into a scattering-medium input for the resonance-line code LaRT (line HeI\_10833): `--format car` (default, uniform grids) writes the all-in-one Cartesian file with `gasDen = n3`, `T = Te`, and zero velocities; `--format amr` writes the generic AMR table with `n3` in the `n_ion` column. `--field` selects a different leaf field for the scatterer. A worked end-to-end case (converted grid + LaRT namelist) is in `examples/hei_metastable/lart/`.

### 3.7 Building AMR grids

`tools/python/AMR_grid/` builds the generic AMR octree file the code reads (`grid_type='amr'`): `AMR_grid.py` (the `AMRGrid` class — uniform, density-gradient, or resolution refinement; `set_density/set_neutral_fraction`; write to

HDF5/FITS/.fits.gz/text with columns x,y,z,level,nH[,metallicity,xHI,ndust]) and make\_amr\_sphere.py (a sphere/shell CLI). AMRGrid carries the same slice/plot\_slice/cutout so a grid can be inspected before a run.

## 4. Worked examples (the regression tests)

Each tests/ directory is self-contained: grid builder or a symlinked grid, the input file, and a check\_\*.py gate script.

| directory    | what it demonstrates / gate                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| g0_gamma     | rate integrals vs. analytic attenuation (bias +0.004%)                                                                                                                                                     |
| g1_stromgren | Strömgren sphere, case B fixed $T_e$ ; AMR $\equiv$ uniform                                                                                                                                                |
| g2a_thermal  | thermal H/He nebula vs. the 1D exact reference                                                                                                                                                             |
| g2_hii       | MOCASSIN Lexington HII20/HII40 benchmarks + line fluxes                                                                                                                                                    |
| g4_refine    | I-front re-refinement vs. native level 7                                                                                                                                                                   |
| g4_tng       | Illustris-TNG post-processing demo (shadows, maps)                                                                                                                                                         |
| d_dusty      | dusty Strömgren; scattering bracket; $T_d$ /IR                                                                                                                                                             |
| peel         | peel-off images vs. the optically thin analytic flux (+0.017%); bin cubes                                                                                                                                  |
| uni_dda      | uniform grid: 'amr' walk $\equiv$ octree (bit-identical); DDA walk to rounding                                                                                                                             |
| pdr          | FUV/PDR switches: carbon-electron shell, photoelectric $T_e$                                                                                                                                               |
| ext_field    | isotropic external field: interior $J = J_{\text{ext}}$ to 0.2% (rec and sph)                                                                                                                              |
| multi_src    | multiple point sources: superposition, spectrum resolution, one-file spectra                                                                                                                               |
| peel_ext     | external-field peel: silhouette flux vs. $JA_{\text{proj}}/d^2$ , chord $\tau$ spectrum                                                                                                                    |
| qmc          | quasi-random launch: Sobol generator vs. scipy + Owen-net balance; the fixed-state variance gate (cell-wise $\Gamma_{\text{HI}}$ gain $29\times$ at $2^{17}$ , $222\times$ at $2^{20}$ )                   |
| hei_meta     | He I $2^3\text{S}$ metastable: OH18 analytic limits, module $n_3$ vs. the Python closed form, $\Phi_3$ from the band tally, and the 10833 Å column/optical-depth diagnostic (conservation + hand integral) |

A minimal thermal + metals + dust input (from d\_dusty):

&parameters

```

par%no_photons = 4.0e7          par%iseed = 1234
par%grid_type = 'amr'          par%amr_file = 'amr_L6.fits'
par%distance_unit = 'pc'       par%dust_model = 'global_dgr'
par%nnu_ion = 32
par%ion_energy_mode = 'grouped' ! dust example; continuous mode currently requires ion_add_dust = .false.
par%eion_min = 13.598          par%eion_max = 100.0
par%tstar = 4.0e4              par%luminosity = 3.178e38
par%xHI_init = 1.0e-3          par%He_abund = 0.1
par%gas_niter = 60             par%gas_tol = 2.0e-3
par%solve_te = .true.          par%atomic_dir = '../data/atomic'
par%use_metals = .true.

```

```

par%ion_add_dust = .true.
par%ion_dust_kext = '../data/kext_albedo_WD_MW_3.1_60_D03.all_2003'
par%cext_dust = 4.868e-22      par%DGR = 1.0
par%out_file = 'run.h5'      par%file_format = 'hdf5'
/

```

## 5. Adding an ion

Adding an ion touches only data:

1. `tools/fitting/fit_cooling.py`: add the ion with a  $T_i$  ladder guessed from its `.elvlc` level energies; run; check the reported max error and the overlay figure.
2. `tools/fitting/fit_nlevel.py`: add the ion with the level count needed for its diagnostics; run.
3. `tools/fitting/make_element_data.py`: add the element to `ELEMENTS` ( $Z$ , stage count) and its thresholds; charge exchange entries where sources exist; run. The CHIANTI `.rrparams/.drparams` types 1–3 are handled (RR/RR2/DR/DR2 lines).
4. Fortran: a `par%abund_<El>` field and one entry in the `species_setup` name/abundance lists (only for a NEW element; new stages of a known element need nothing).
5. Verify: extend `verify_nlevel_pyneb.py` with a diagnostic ratio where PyNeb has data; run a smoke test and check the new lines in `_lines.txt`.

Data caveats met so far: Ar II has no CHIANTI v11 level/collision data (stage tracked without lines); Fe III collision strengths differ from PyNeb's BB14 by up to  $\sim 16\%$  in 4658/4702 (known spread for iron); charge-exchange stage labels differ between the Huang (2023) and MOCASSIN transcriptions for Mg/Fe (the Huang labels are adopted; see `make_element_data.py`).

## 6. Practical notes

- **Bins**: 32 bins give  $\sim 2\%$  rate-integral discretization; use 64 for sub-percent work. The `add_fuv` extension carries its own bins below `eion_min`, so the ionizing resolution is unaffected.
- **FUV and the PDR**: without `add_fuv` there is no radiation beyond the I-front and the metals there recombine toward neutral; with it, FUV penetrates the I-front and photoionizes the metal stages whose thresholds lie below 13.6 eV (C I 11.3, S I 10.4, Si I 8.2, Fe I 7.9, Mg I 7.6 eV, ...), producing the PDR-side C II and Mg II. In the FUV band the gas opacity is exactly this low-threshold metal photoionization (`ion_metal_abs`, on by default; H and He do not absorb below 13.6 eV), so the FUV is attenuated by that photoionization together with dust — only where those metals are already ionized does dust extinction alone remain. The metal stage fractions iterate with this opacity self-consistently; the corresponding metal photoheating and the grain photoelectric heating are not in the thermal balance, so PDR-zone temperatures are indicative only.

- **Convergence:** use `conv_crit = 'vol'` for production — the cell-max criterion stalls at the front-cell noise floor (and, with `add_fuv`, at no-heating cells where  $T_e$  pins to `te_min`). Both measures are printed each iteration. The explicit diffuse field smooths the front and converges far better.
- **Starts:** prefer an ionized initial state; a neutral start advances the front  $\sim$ one cell per iteration.
- **text tables:** must cover the band (the astro dust table stops at 912 Å; use the D03 table for EUV work). Tables are sorted on read (the D03 file has an out-of-order seam).
- **Re-refinement:** one event per run; the old shared-memory windows are reclaimed only at exit.
- **SEDust:** `sed_workdir` must point at a SEDust `sed/` tree; the exact stochastic solve costs  $\sim 0.05\text{--}0.1$  s for each gas leaf (distributed over ranks).